

CSE/ISyE 6740
Computational Data Analysis

Gaussian Process Regression

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Kai Wang, Assistant Professor in Computational Science and Engineering

kwang692@gatech.edu

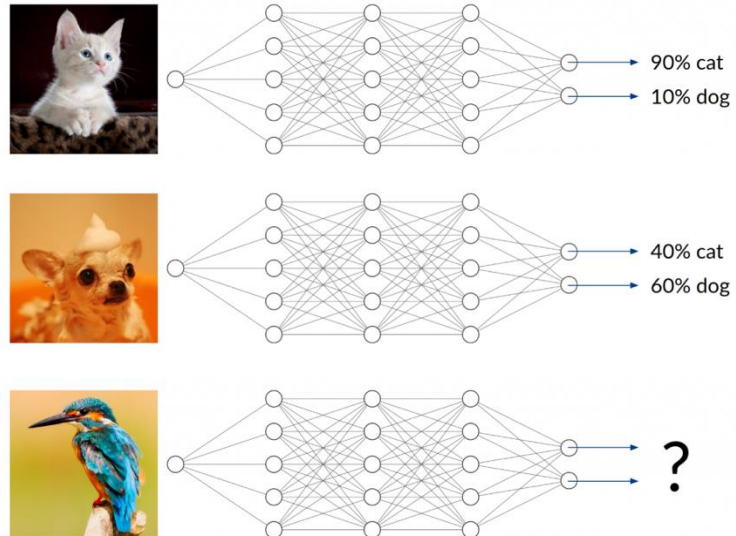
Outline

- Supervised Learning
 - Kernel methods
 - Uncertainty quantification / probabilistic regression models
 - Bayesian regression (linear)
 - Gaussian process regression (non-linear)

Uncertainty Quantification

Probabilistic Meaning of Models

- Can we estimate the uncertainty of the predictions?
 - How confident is the prediction?



- Can the model have some probabilistic meaning?
 - Prior / posterior?

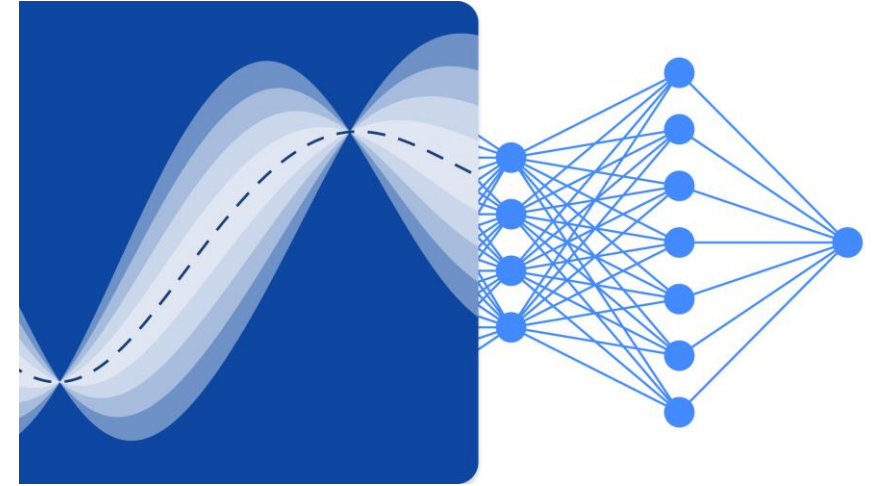
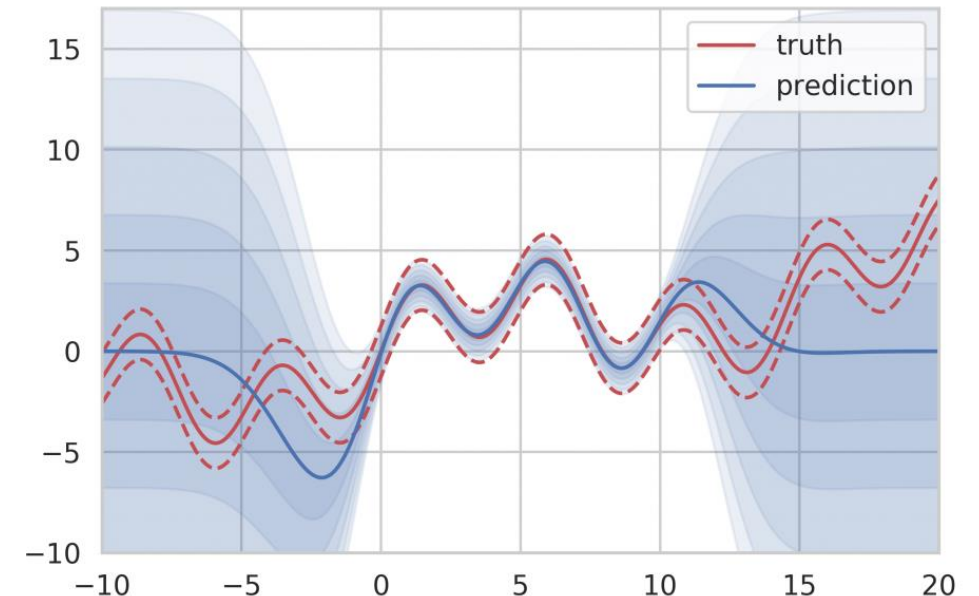


Image credit: <https://www.inovex.de/de/blog/uncertainty-quantification-deep-learning/>



Bayesian Linear Model

Bayesian Linear Model

- **Linear model:** $y(x) = \theta^\top x + \epsilon$
 - $x \in \mathbb{R}^d$ is the input feature and $\theta \in \mathbb{R}^d$ is the weight vector, $\epsilon \sim N(0, \sigma^2)$
 - $\log p(D|\theta) = \sum_{i=1}^n \log N(y^i - \theta^\top x^i, \sigma^2) = \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^i - \theta^\top x^i)^2}{2\sigma^2}\right)$
$$= \text{const} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y^i - \theta^\top x^i)^2$$
$$= \text{const} - \frac{1}{2\sigma^2} \|y - X^\top \theta\|^2$$
- **Bayesian:** we additionally assume $\theta \sim N(0, S)$

Bayesian Linear Model

- We can therefore compute

$$p(\theta|D) = \frac{p(D|\theta)p(\theta)}{p(D)} \propto p(D|\theta)p(\theta)$$

- This Bayesian assumption on $p(\theta)$ will also slightly change how we make predictions:

$$p(y|x, D) = \int p(\theta|D) p(y|x, \theta) d\theta$$

- In the non-Bayesian case, we use $y = \theta^{*\top} x$ once we find the optimal $\theta^* = \operatorname{argmax}_{\theta} p(D|\theta)$.
- **Linear regression** only considers the best fit $\theta^* = \operatorname{argmax}_{\theta} p(D|\theta)$
- **Bayesian linear regression** considers all plausible explanations for how the data were generated $\theta \sim N(0, S)$.

Bayesian Linear Model Training

- $p(\theta|D) \propto p(D|\theta)p(\theta)$
- $\log p(\theta|D) = \log p(\theta) + \log p(D|\theta) + \text{const}$
$$= -\frac{1}{2}\theta^\top S^{-1}\theta - \frac{1}{2\sigma^2}\|y - X\theta\|^2 + \text{const}$$
$$= -\frac{1}{2}\theta^\top S^{-1}\theta - \frac{1}{2\sigma^2}(y^\top y - 2\theta^\top X^\top y + \theta^\top X^\top X\theta) + \text{const}$$
$$= -\frac{1}{2}(\theta - \mu)^\top \Sigma^{-1}(\theta - \mu) + \text{const}$$

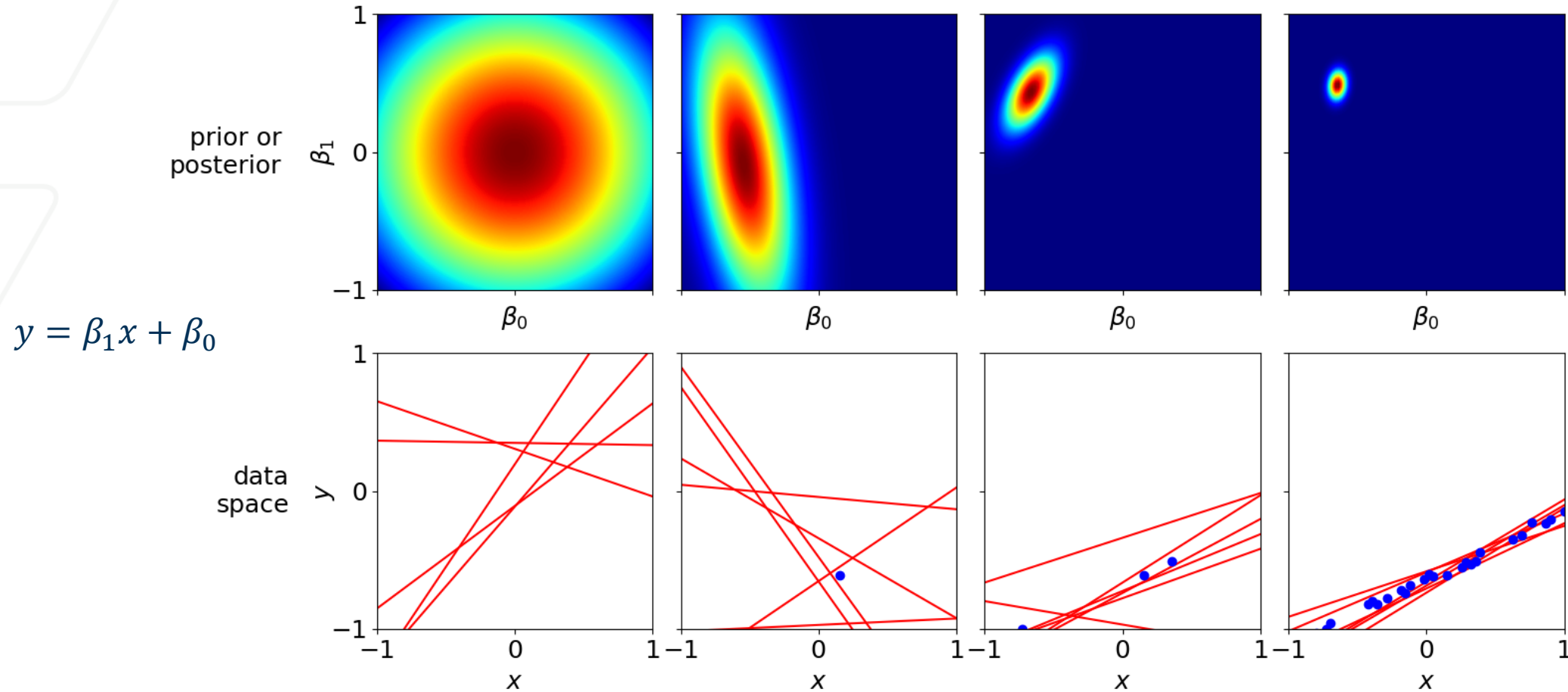
Where $\mu = \frac{1}{\sigma^2} \Sigma X^\top y$

And $\Sigma^{-1} = \frac{1}{\sigma^2} X^\top X + S^{-1}$

Therefore, $p(\theta|D) = N(\mu, \Sigma)$ (multi-variate Gaussian distribution)

This looks terrifying
but this is just
complete the square.

Bayesian Linear Model Example



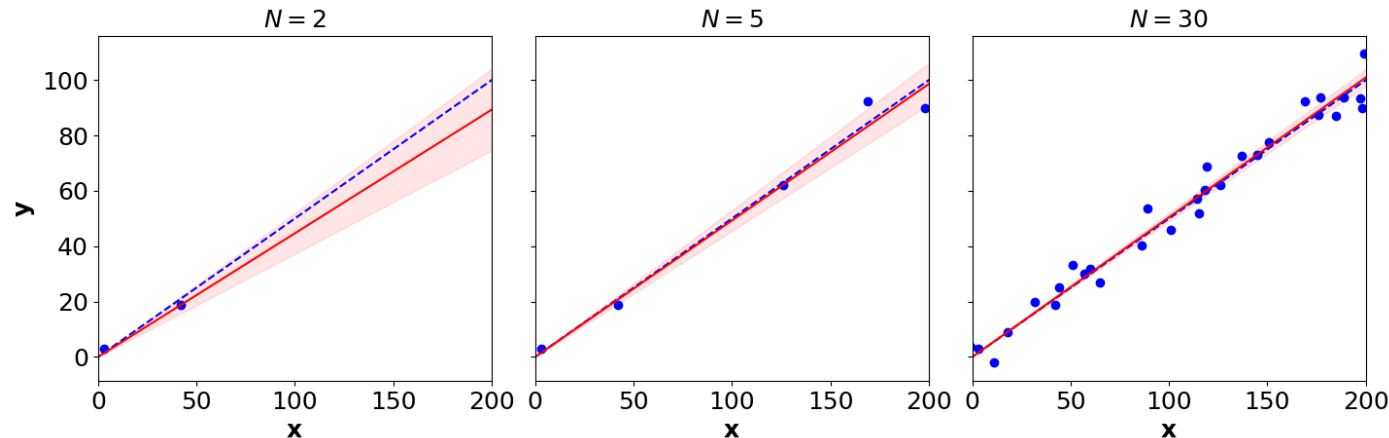
Bayesian Linear Model Inference

- Given $p(\theta|D) = N(\mu, \Sigma)$ with $\mu = \frac{1}{\sigma^2} \Sigma X y$ and $\Sigma^{-1} = \frac{1}{\sigma^2} X X^\top + S^{-1}$, inference of a new x can be done by:

$$p(y|x, D) = \int \underbrace{p(y|x, \theta)}_{y \sim N(\theta^\top x, \sigma^2)} \underbrace{p(\theta|D)}_{\theta \sim N(\mu, \Sigma)} d\theta$$

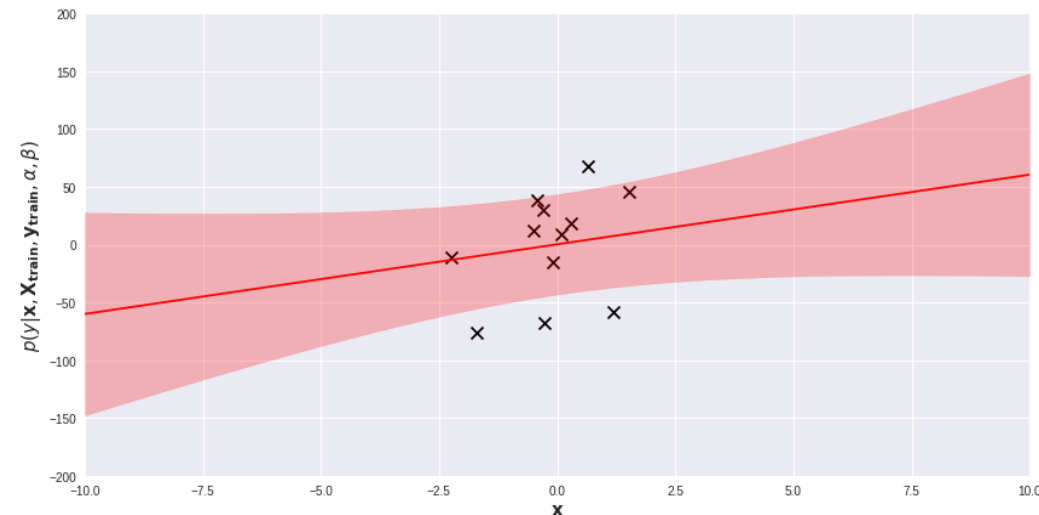
By the linear combination rules of Gaussian random variables, $y|x, D$ is also a Gaussian distribution $N(\mu_{pred}, \sigma_{pred}^2)$ with parameters:

$$\mu_{pred} = \mu^\top x, \quad \sigma_{pred}^2 = x^\top \Sigma x + \sigma^2$$

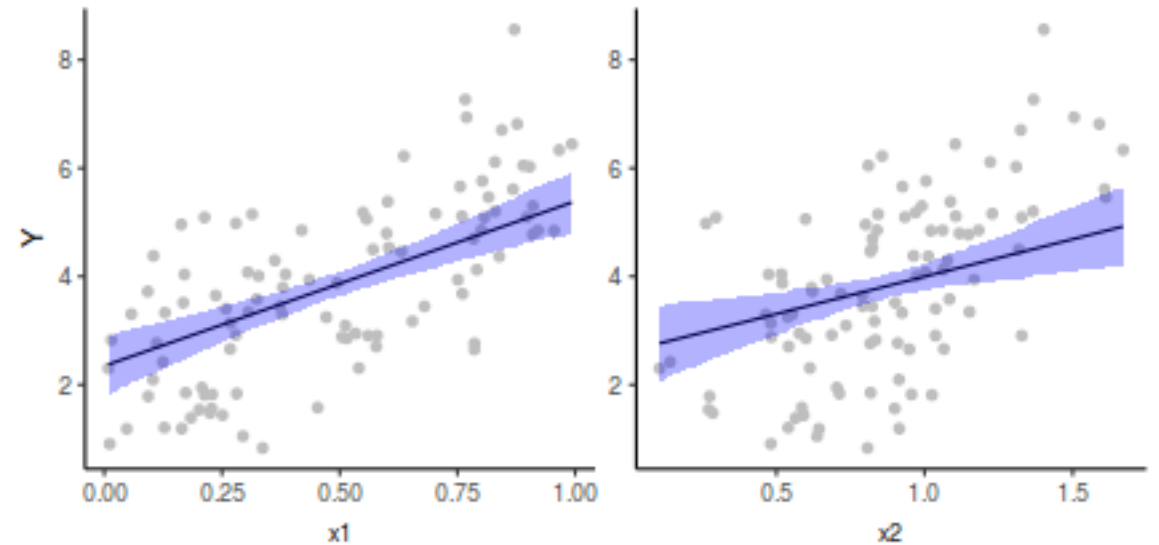
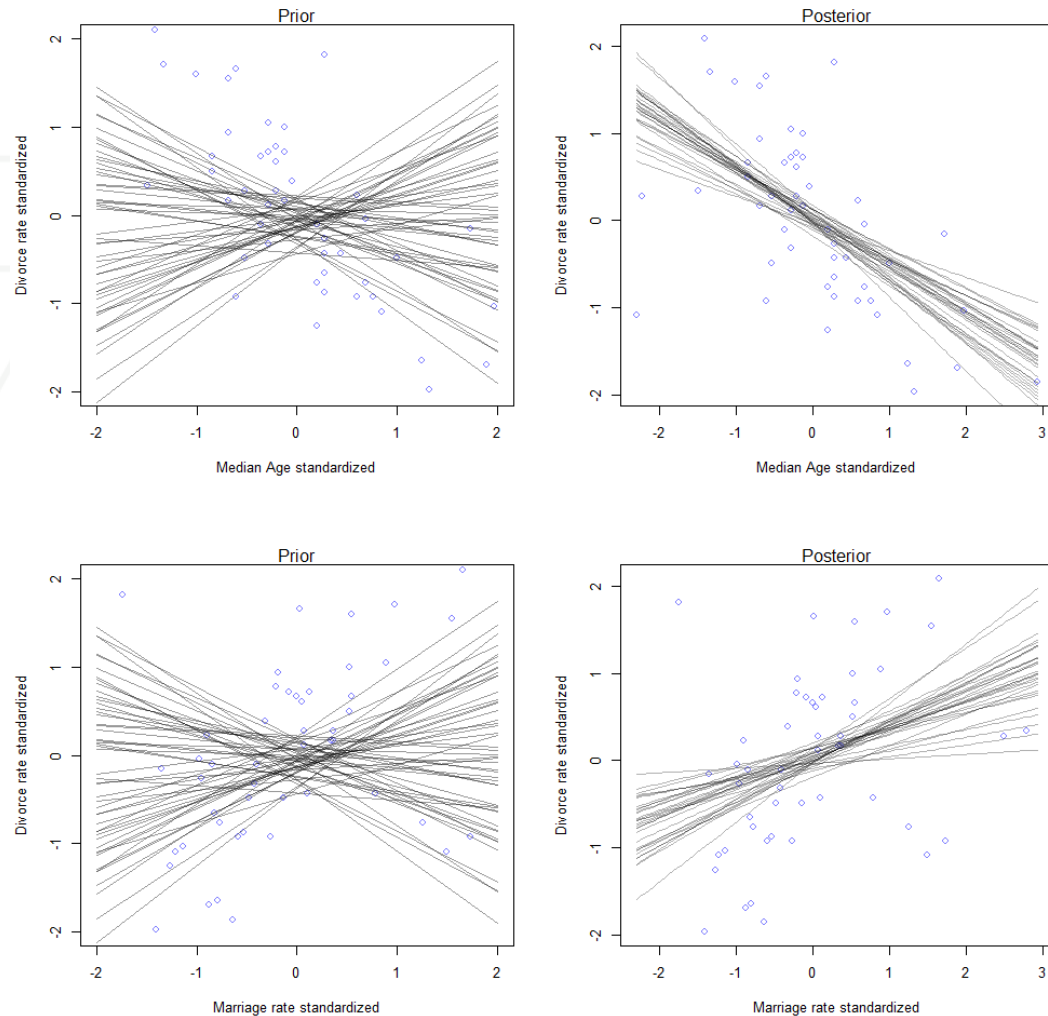


Bayesian Linear Model Summary

- $\mu = \frac{1}{\sigma^2} \Sigma X^\top y = (X^\top X + \sigma^2 S^{-1})^{-1} X^\top y$
- $\Sigma^{-1} = \frac{1}{\sigma^2} X^\top X + S^{-1}$
- $\mu_{pred} = \mu^\top x = x^\top \mu = x^\top (X^\top X + \sigma^2 S^{-1})^{-1} X^\top y$
- $\sigma_{pred}^2 = x^\top \Sigma x + \sigma^2 = \sigma^2 x^\top (X^\top X + \sigma^2 S^{-1})^{-1} x + \sigma^2$



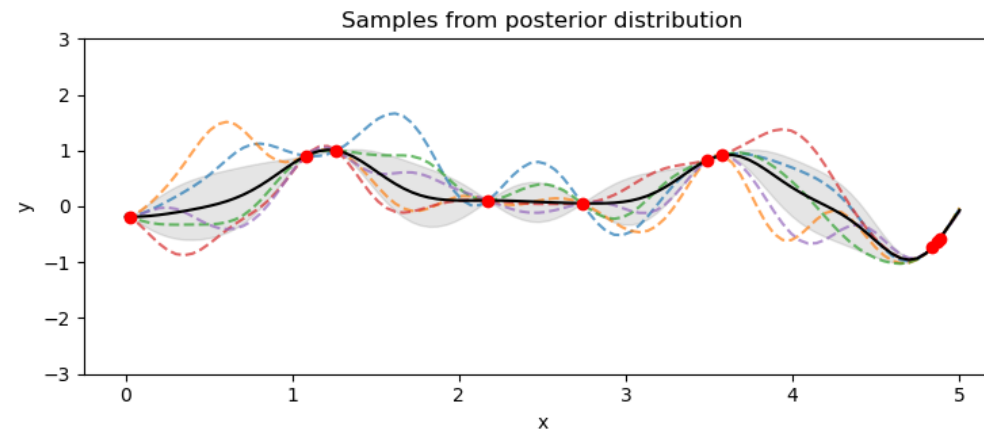
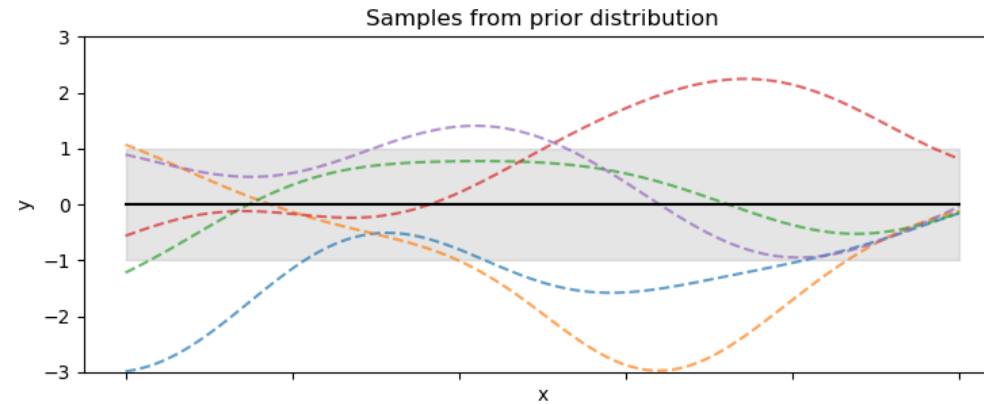
Bayesian Linear Model Prediction Example



Gaussian Process Regression

Can We Apply Kernel Tricks Here?

- Yes, and it is exactly the **Gaussian process regression**!!



Bayesian Linear Model Prediction

- $\mu = \frac{1}{\sigma^2} \Sigma X^\top y = (X^\top X + \sigma^2 S^{-1})^{-1} X^\top y$
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- $\sigma_{pred}^2 = x^\top \Sigma x + \sigma^2 = \sigma^2 x^\top (X^\top X + \sigma^2 S^{-1})^{-1} x + \sigma^2$

Kernel Tricks on Bayesian Linear Regression

- $\mu = \frac{1}{\sigma^2} \Sigma \tilde{X}^\top y = (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \tilde{X}^\top y$
- $\Sigma^{-1} = \frac{1}{\sigma^2} \tilde{X}^\top \tilde{X} + I$

Assume $S = I$ for simplicity

- $\mu_{pred} = \mu^\top \phi(x) = \phi(x)^\top \mu = \phi(x)^\top (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \tilde{X}^\top y$
- $\sigma_{pred}^2 = \phi(x)^\top \Sigma \phi(x) + \sigma^2 = \sigma^2 \phi(x)^\top (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \phi(x) + \sigma^2$

Gaussian Process Regression

Matrix inversion
lemma

$$\begin{aligned}\bullet \mu_{pred} &= \mu^\top \phi(x) = \phi(x)^\top \mu = \phi(x)^\top (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \tilde{X}^\top y \\ &= \phi(x)^\top \tilde{X}^\top (\tilde{X} \tilde{X}^\top + \sigma^2 I)^{-1} y \\ &= k_x^\top (K + \sigma^2 I)^{-1} y\end{aligned}$$

$$\bullet \sigma_{pred}^2 = \phi(x)^\top \Sigma \phi(x) + \sigma^2 = \sigma^2 \phi(x)^\top (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \phi(x) + \sigma^2$$

Woodbury identify

$$(I + UV)^{-1} = I - U(I + VU)^{-1}V$$

Woodbury matrix identity

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Article Talk

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From Wikipedia, the free encyclopedia

In [mathematics](#), specifically [linear algebra](#), the **Woodbury matrix identity** – named after [Max A. Woodbury](#)^{[1][2]} – says that the inverse of a rank-*k* correction of some [matrix](#) can be computed by doing a rank-*k* correction to the inverse of the original matrix. Alternative names for this formula are the **matrix inversion lemma**, **Sherman–Morrison–Woodbury formula** or just **Woodbury formula**. However, the identity appeared in several papers before the Woodbury report.^{[3][4]}

The Woodbury matrix identity is^[5]

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1},$$

Gaussian Process Regression

- $\sigma_{\text{pred}}^2 = \phi(x)^\top \Sigma \phi(x) + \sigma^2 = \sigma^2 \phi(x)^\top (\tilde{X}^\top \tilde{X} + \sigma^2 I)^{-1} \phi(x) + \sigma^2$

$$= \phi(x)^\top \left(\underbrace{\frac{1}{\sigma^2} \tilde{X}^\top \tilde{X}} + I \right)^{-1} \phi(x) + \sigma^2$$

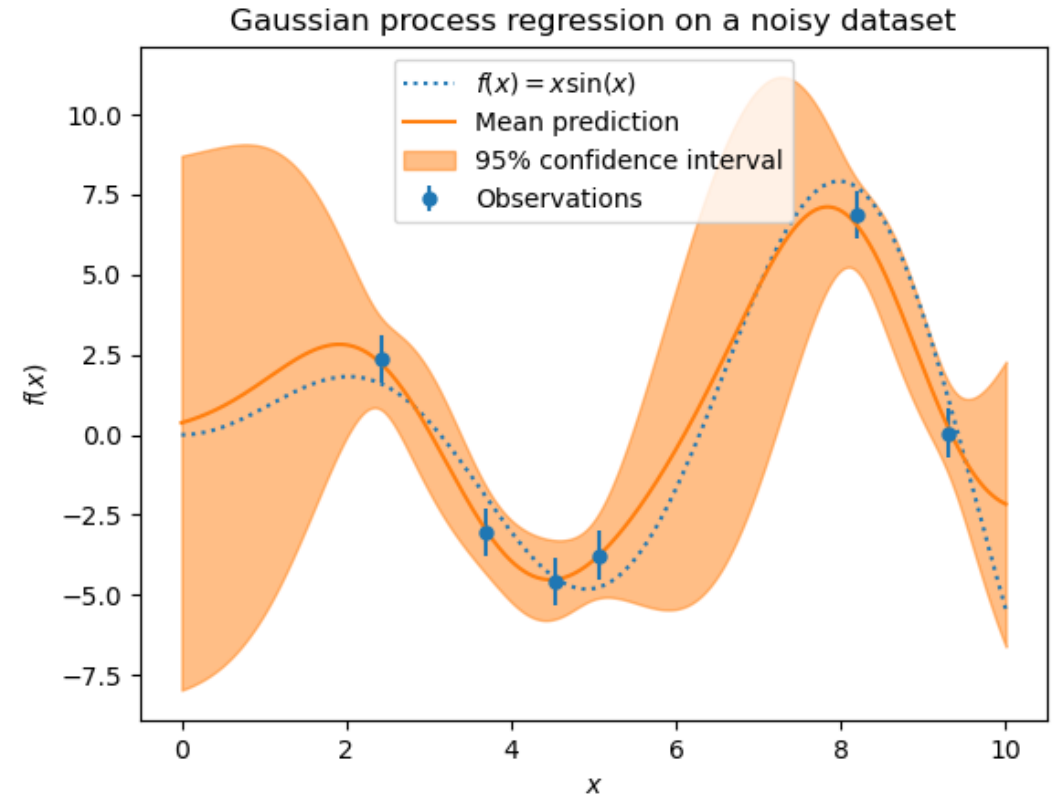
$$U = \frac{1}{\sigma} \tilde{X}^\top \quad V = \frac{1}{\sigma} \tilde{X}$$

$$\begin{aligned} &= \phi(x)^\top (I + UV)^{-1} \phi(x) + \sigma^2 \\ &= \phi(x)^\top (I - U(I + VU)^{-1} V^\top) \phi(x) + \sigma^2 \\ &= \phi(x)^\top \phi(x) - \phi(x)^\top \tilde{X}^\top (K + \sigma^2)^{-1} \tilde{X} \phi(x) + \sigma^2 \end{aligned}$$

$$= k_{xx} - k_x^\top (K + \sigma^2)^{-1} k_x + \sigma^2$$

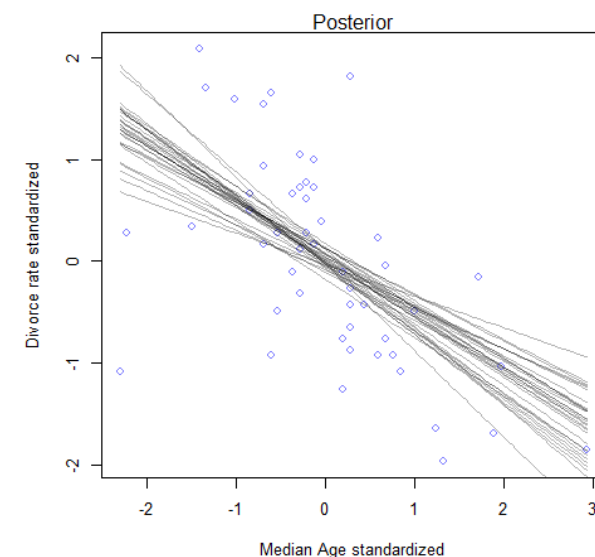
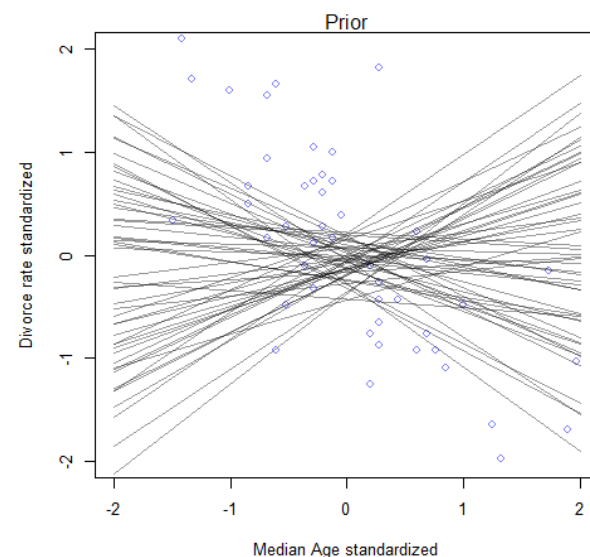
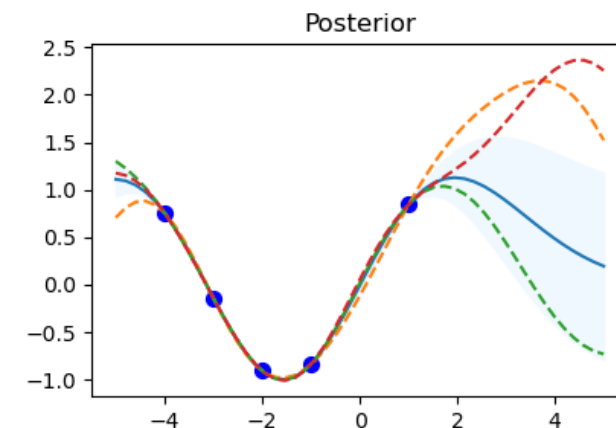
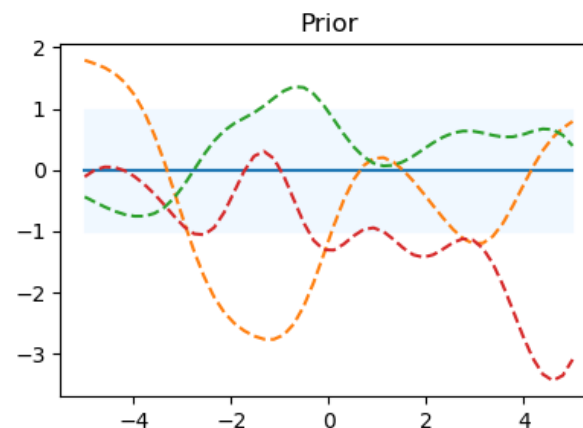
What Does That Mean???

- $\mu_{pred} = k_x^\top (K + \sigma^2 I)^{-1} y$
- $\sigma_{pred}^2 = k_{xx} - k_x^\top (K + \sigma^2)^{-1} k_x + \sigma^2$
- For any new point x , a kernel function k , and a dataset $D = (x^{(i)}, y^{(j)})$, we just need to compute:
 - $k_{xx} = k(x, x)$
 - $k_x = [k(x, x^{(i)})]_i \in \mathbb{R}^n$
 - $K = [k(x^{(i)}, x^{(j)})]_{ij} \in \mathbb{R}^{n \times n}$
- Then we can plot its posterior belief with mean and variance at x



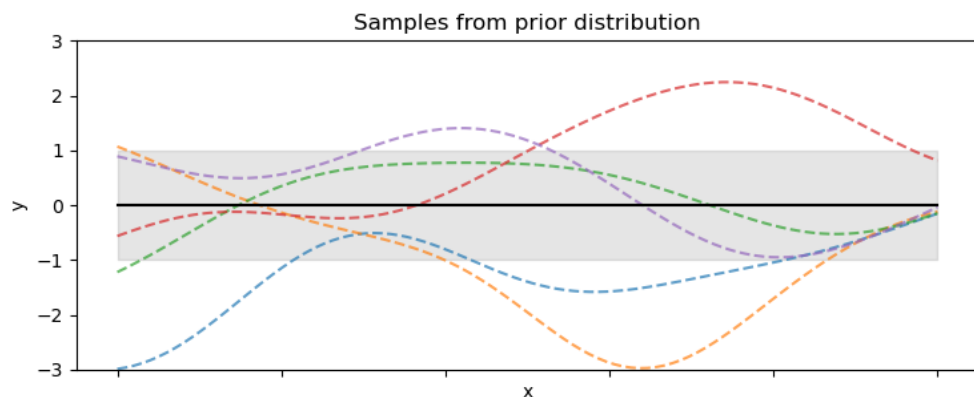
Gaussian Process Regression

- We can generate posterior distribution of each point x .
- This is extremely useful when you want to measure the uncertainty of the prediction.
- You can precompute the matrix inversion to speed up computation.

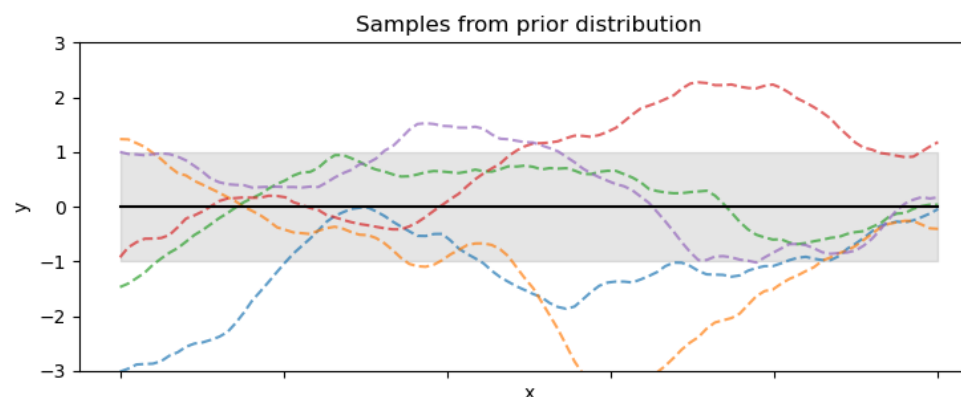


More Gaussian Process Regression

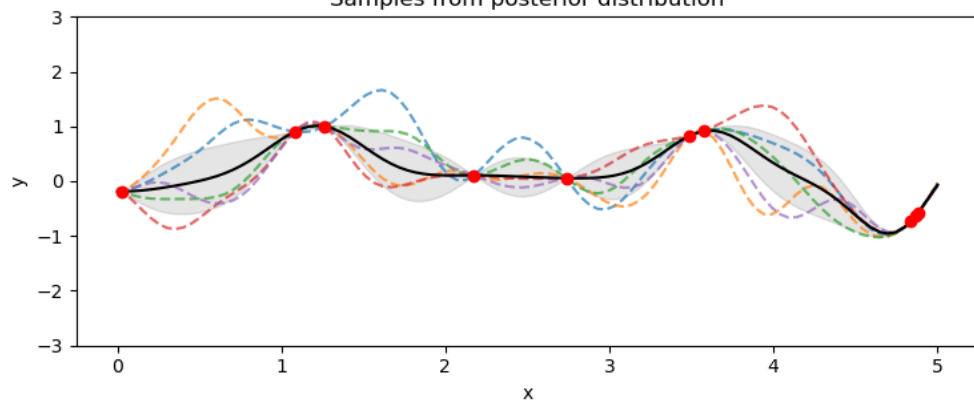
Radial Basis Function kernel



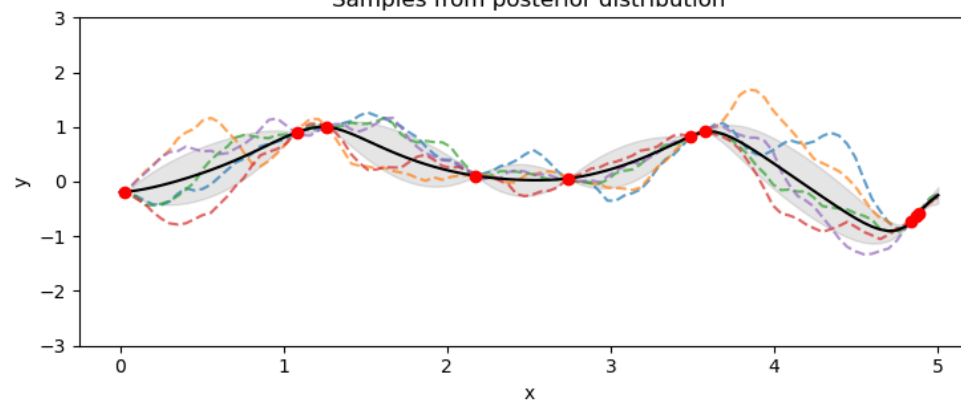
Matérn kernel



Samples from posterior distribution



Samples from posterior distribution



Summary

- Kernel tricks!!
- From linear to nonlinear
- We also learn how to produce uncertainty by generating a posterior distribution

